

AdaBoost Algorithm

Mathematical Formulation for Binary Classification

1. Problem Setup and Notation

Let the training dataset be defined as:

$$D = \{ (x_i, y_i) \} \text{ for } i = 1, 2, \dots, N$$

where $x_i \in \mathbb{R}^p$ represents the feature vector, $y_i \in \{-1, +1\}$ represents the binary class label, and N denotes the total number of training observations.

At initialization, each observation is assigned equal weight:

$$w_i(1) = 1 / N \text{ for all } i$$

2. Weighted Classification Error

At iteration m , a weak learner $h_m(x)$ is trained using the current weight distribution.

The weighted classification error is defined as:

$$\varepsilon_m = \sum_i w_i(m) I(y_i \neq h_m(x_i))$$

where $I(\cdot)$ denotes the indicator function and $w_i(m)$ represents the weight of observation i at iteration m . Further, $h_m(x_i)$ denotes the prediction of the m -th weak classifier for observation x_i .

For boosting to proceed effectively, the weak learner must satisfy:

$$\varepsilon_m < 0.5$$

3. Weak Learner Coefficient

The contribution of the m -th weak learner is determined by:

$$\alpha_m = \frac{1}{2} \ln \left[\frac{(1 - \varepsilon_m)}{\varepsilon_m} \right]$$

$\nu \in (0,1]$ is the learning rate (shrinkage parameter)

A smaller error ε_m results in a larger value of α_m , thereby giving greater influence to stronger weak learners.

4. Weight Update Rule

After computing α_m , the sample weights are updated according to:

$$w_i(m+1) = w_i(m) \exp(-\alpha_m y_i h_m(x_i))$$

Misclassified observations receive increased weights, while correctly classified observations receive decreased weights.

5. Normalization

To maintain a valid probability distribution, weights are normalized as follows:

$$w_i(m+1) = w_i(m+1) / \sum_j w_j(m+1)$$

This ensures that the sum of weights at iteration $m+1$ equals one.

6. Final Strong Classifier

After M boosting iterations, the strong classifier is defined as:

$$F(x) = \sum_m \alpha_m h_m(x)$$

The final prediction rule is:

$$\hat{y} = \text{sign}(F(x))$$

7. Exponential Loss Interpretation

AdaBoost can be interpreted as minimizing the exponential loss function:

$$L = \sum_i \exp(-y_i F(x_i))$$

This interpretation connects AdaBoost to statistical learning theory and stagewise additive modeling.